What do spectral criteria of quantum feature-map kernels actually predict? Classical simulability, generalization advantage, and the role of task difficulty

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Quantum feature maps have been proposed as a route to machine-learning advantage, yet two separate questions are often conflated: whether a quantum kernel is *classically simulable*, and whether it yields a real *generalization advantage* over strong classical baselines. We study what spectral criteria of the quantum kernel matrix — effective rank, spectral entropy, concentration index, gap ratio — can and cannot predict, using a self-contained state-vector simulator (no external quantum libraries), four feature-map families and six controlled-difficulty tasks, and tuned classical baselines (radial basis function, RBF, with validation-based kernel width, random Fourier features, linear and polynomial kernels). Across 781 configurations we find that spectral criteria reliably predict classical simulability (top-5 energy $\rho = -0.80$, $p = 1.2 \times 10^{-15}$; spectral entropy $\rho = +0.67$, $p = 1.0 \times 10^{-9}$; Benjamini–Hochberg corrected), but they do not predict quantum advantage: the global correlation is indistinguishable from zero ($|\rho| \leq 0.07$, $p \geq 0.31$), and its sign is unstable across tasks ($\rho = +0.38$ in a single-task setting, $\rho = -0.63$ within the cross-term task, null elsewhere). The dominant predictor of advantage is the saturation of classical baselines: advantage correlates strongly and negatively with the best classical test accuracy ($\rho = -0.65$, $p = 2 \times 10^{-27}$), and positive advantage occurs almost exclusively where classical baselines remain near chance (39% at 0.55, 0% at 0.86). The results hold at 8–16 qubits, on four real datasets (0/48 positive advantage), and under depolarizing noise, while amplitude damping reveals a boundary: spectral collapse stops indicating classical approximability ($\rho = +0.04$ versus $+0.35$). Kernel-target alignment, a popular matching criterion, is also absorbed by task difficulty ($\rho = -0.42$ globally, $\rho = +0.001$ after residualization). These results draw a sharp boundary for spectral pre-screening and provide evaluation guidelines for quantum kernel studies.

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1 Introduction

The promise of quantum machine learning rests on a single question: does embedding classical data into an exponentially large Hilbert space give a model an ability that classical models lack? In the kernel framework this question becomes sharp. A quantum feature map $\phi(x)$ maps data into the state space of n qubits, and the resulting fidelity kernel $k(x, y) = |\langle\phi(x)|\phi(y)\rangle|^2$ is used exactly like a classical kernel in a kernel machine [1, 2].

Two lines of recent work have questioned the optimistic picture. On one side, quantum kernels can suffer *exponential concentration*: under broad conditions the kernel values cluster so tightly that the kernel matrix loses all discriminative power [3]. On the other, a growing body of work shows that many "quantum" models can be replaced by classical surrogates—the dequantization program [4, 5], including results on quantum learning models that are classically simulable. More recently, random Fourier features have been proposed as a canonical classical surrogate that can be matched against a quantum model [6].

This has motivated a practical ambition: before paying the cost of quantum computation or training, predict *whether* a given feature map will be effective. A natural candidate is the spectrum of the quantum kernel matrix itself. Cheap spectral statistics—effective rank, spectral entropy, concentration—are easy to compute from small samples, and have been used both as diagnostics of concentration and as proxies for classical simulability.

Related work approaches this ambition from several directions. Kübler et al. [7] analyzed the inductive bias of quantum kernels theoretically and argued that advantage can be expected when the reproducing-kernel Hilbert space is low dimensional but contains functions that are hard to compute classically. Heyraud et al. [8] introduced the effective kernel rank as a figure of merit for noisy quantum kernels, and Zendejas-Morales et al. [9] use effective rank as a concentration diagnostic in mitigation studies. On the empirical side, Kakavand et al. [10] benchmarked quantum against classical kernels on nine datasets with 970 experiments and found that dataset choice dominates performance variance while kernel type contributes little, observing also that typical quantum feature maps produce spectra that are either too flat or too concentrated relative to RBF. Appel [11] proposed the correlation fractal dimension as an a-priori qubit budget for angle-encoded kernels, and Åsgrim and Markidis [12] showed that quantum kernels can be represented as spectral tensor networks whose compressibility diagnoses classical tractability. Our contribution differs in one respect that these studies do not address directly: we ask which of the two questions spectral criteria can answer *at all*, separating predictions about simulability from predictions about advantage, and we expose the task-dependence of advantage through a classical-saturation predictor.

Yet the two questions such criteria are asked to answer are logically different. *Classical simulability* is a property of the kernel function alone: can a classical kernel family approximate it? *Generalization advantage* is a property of a triple: kernel, task, and baseline. A kernel can be simulable without the classical surrogate winning on a particular task, and an advantage can be absent even when the kernel is highly expressive, simply because the task is easy.

In this work we ask: *what do spectral criteria actually predict?* We adopt a task-resolved numerical protocol, using only a self-contained state-vector simulator, four feature-map families with tunable design knobs, controlled task structures, and tuned classical baselines. The protocol lets us separate the two questions empirically rather than theoretically.

1.1 Contributions

Assumptions. All results below are numerical and obtained with exact state-vector simulation of $n \leq 5$ qubit circuits on $N \leq 120$ samples, without noise. Four feature-map families (angle, IQP, variational, high-dimensional) are studied under exactly reproducible seeds; six tasks of controlled structure and difficulty (synthetic plus a real dataset) probe the kernel–task interaction. Advantage is measured against tuned classical baselines (SVM and kernel methods with validation-based hyperparameters).

Main results.

1. Spectral criteria reliably predict *classical simulability* as measured by RFF and Nyström approximation error; this is a clean, task-independent signal, with top-5 energy at $\rho = -0.80$ ($p = 1.2 \times 10^{-15}$) (Sec. 4.1).

2. Spectral criteria do *not* predict quantum advantage globally. Across six tasks, the correlation between effective rank and advantage is indistinguishable from zero ($\rho = -0.02$, $p = 0.76$), and an apparently strong correlation in a single-task setting ($\rho = +0.38$) reverses sign within a different task (Sec. 4.2).

3. The dominant predictor of advantage is the *saturation of classical baselines* on the task: the best classical test accuracy correlates with advantage at $\rho = -0.65$ ($p = 1.8 \times 10^{-27}$), and the fraction of positive-advantage configurations falls from 39% to 0% as the classical baseline rises from 0.55 to 0.86 (Sec. 4.3).

4. The frequency scale of the feature map is the decisive design knob for spectral spread and simulability ($\rho = +0.985$, $p = 3 \times 10^{-34}$), while entanglement strength is not (Sec. 4.4); spectral criteria are also strongly data-dependent, in ways not explained by data intrinsic dimension (Sec. 4.5).

5. The central relations are robust to scaling, real data, and noise: the simulability relation strengthens at 8–12 qubits ($\rho = +0.80$), positive advantage is never observed on four real datasets (0/48), and all conclusions survive 30% depolarizing noise (Sec. 4.6). Two natural matching criteria—spectral spread ratio and kernel-target alignment—fail to predict advantage beyond classical difficulty, with alignment's apparent power entirely absorbed by it (Sec. 4.7).

2 Definitions and background

2.1 Quantum feature maps and fidelity kernels

A quantum feature map $\phi : \mathbb{R}^d \to \mathcal{H}$ maps a classical feature vector x to a normalized state $\phi(x) = U(x)|0 \cdots 0\rangle$, where $U(x)$ is a data-encoding circuit. The induced kernel is $k(x, y) = |\langle\phi(x)|\phi(y)\rangle|^2$. Given a dataset $X = \{x_i\}_{i=1}^N$, the kernel matrix $K_{ij} = k(x_i, x_j)$ feeds classical kernel methods. Throughout we study small systems ($n \leq 5$ qubits, $N \leq 120$), where the state vector has at most 32 amplitudes and exact simulation is trivial on a CPU.

2.2 Spectral criteria

Let $\lambda_1 \geq \cdots \geq \lambda_N \geq 0$ be the eigenvalues of K, and $p_i = \lambda_i / \sum_j \lambda_j$. We use the following criteria, all computable in $O(N^3)$:

- **Effective rank** (participation ratio): $r_{\text{eff}} = (\sum_i \lambda_i)^2 / \sum_i \lambda_i^2$, ranging from 1 (perfectly concentrated) to $\text{rank}(K)$.
- **Spectral entropy**: $H = -\sum_i p_i \log_2 p_i$, with maximum $\log_2 N$.
- **Concentration index**: $c = 1 - r_{\text{eff}}/N$.
- **Gap ratio**: $\lambda_1/(\lambda_1 + \lambda_2)$.
- **Top-k energy**: $\sum_{i \leq k} \lambda_i / \sum_i \lambda_i$ ($k = 5$).

2.3 Classical simulability metrics

Given K and the original data matrix X, we measure approximability by two classical families. *Random Fourier features* (RFF) [13] approximate a translation-invariant kernel $k(x - y)$ by random features $z(x)$ with $z(x) \cdot z(y) \approx k(x - y)$; we fit the RFF kernel to the data and measure $\|K_{\text{RFF}} - K\|_F / \|K\|_F$. *Nyström approximation* with m landmarks measures low-rank approximability. Small errors mean the kernel is easy to reproduce classically. Note that RFF approximates only translation-invariant kernels; this restriction becomes important in Sec. 4.4.

2.4 Quantum advantage and classical baselines

For a task with training/test split, define

$$\text{Adv} \;=\; \text{acc}_Q \;-\; \max_{\mathcal{B}} \text{acc}_B, \tag{1}$$

where acc_Q is the test accuracy of a kernel-SVM using the quantum kernel, and the maximum is over a strong classical baseline family $\mathcal{B}$: RBF kernel with the width selected on a validation split, RFF with the same width, linear, and degree-3 polynomial kernels. Using tuned baselines is critical; an untuned baseline would inflate apparent advantage.

2.5 Tasks

Tasks have controlled difficulty and structure: (i) *harmonic* (sum of fixed-frequency sinusoids), (ii) *periodic-coupled* (coupling of periodic terms in two variables), (iii) *cross-high* (pairwise products and cubic terms), (iv) *quantum-friendly* (frequency-doubling sinusoids, matched to harmonic encoding), (v) *noisy-harmonic* (harmonic with 20% label noise), and (vi) *breast-cancer* (real dataset, PCA to d dimensions, standardized). Data are standardized in every case. Each task's classical difficulty is measured by the best baseline accuracy.

3 Methods

3.1 Simulator and feature maps

All simulations use our own state-vector simulator written in NumPy, with no dependence on external quantum libraries; this keeps every result reproducible with only NumPy, SciPy, scikit-learn, and matplotlib. Unit tests verify unitarity of all gates, the fidelity kernel on orthogonal states, and that all feature-map encodings produce normalized states; these tests guard against coding errors in the physics, an issue that we found to matter in practice (see Sec. 4.4 for a discussion of encoding-construction pitfalls).

Four feature-map families are implemented: *angle* (per-qubit R_y rotations, no entanglement), *IQP* (data-dependent R_z rotations preceded by Hadamard layers, followed by a CNOT ladder and quadratic phases), *variational* (angle encoding with repeated entangling+re-encoding layers), and *high-dimensional* (per-qubit multi-frequency rotations with controlled phase entanglement). Design knobs: circuit depth, entanglement strength $e \in [0, 1]$, and frequency scale s.

3.2 Experimental protocol

- **exp01 (simulability)**: 4 maps $\times$ 3 depths $\times$ 4 seeds = 64 configurations on a Gaussian dataset ($N = 40$, $d = 4$); measure all spectral criteria and RFF/Nyström errors.
- **exp02 (single-task advantage)**: 4 maps $\times$ 3 depths $\times$ 6 seeds = 72 configurations on one synthetic task; measures advantage, revealing a seemingly strong spectral-advantage correlation.
- **exp03 (task-resolved advantage)**: 6 tasks $\times$ 12 map variants $\times$ 3 seeds = 216 configurations; measures advantage against tuned baselines, plus stratified analysis. Supplementary analysis measures advantage vs. classical difficulty.
- **exp04 (knobs)**: 114 configurations sweeping frequency scale, entanglement, and depth, measuring spectral criteria and RFF error.
- **exp05 (data dependence)**: 5 map configs $\times$ 6 data distributions $\times$ 4 seeds = 120 configurations; measures the variation of effective rank across data distributions and its relation to PCA intrinsic dimension.
- **exp06 (scaling)**: 51 configurations at 8, 12, and 16 qubits ($N = 250$, two tasks); checks whether the central relations survive larger systems.
- **exp07 (real data)**: 48 configurations on four standard datasets (iris, wine, breast-cancer, digits with PCA to ≤ 8 features, 8 qubits).

- **exp08 (noise)**: 36 configurations under depolarizing noise ($p \in \{0, 0.02, 0.1, 0.3\}$, density-matrix simulation at 5 qubits).
- **exp09 (matching criteria)**: reconstruction of all 216 exp03 kernels to test two task–map matching criteria: the spectral spread ratio $M_1 = \mathrm{rank}_Q/\mathrm{rank}_{\mathrm{RBF}}$, and the centered kernel-target alignment $\mathrm{TA}(K, yy^\top)$.

3.3 Statistics

Associations are measured with Spearman rank correlation. Group comparisons use Mann–Whitney U tests with Cliff's delta effect sizes and threshold scans. All multiple-comparison corrections use the Benjamini–Hochberg FDR procedure; corrected values are reported as q. Significance is claimed only when $q < 0.05$.

Two caveats about the statistics are stated explicitly. First, within each (task, seed) block the twelve feature-map variants share the same data, so configuration-level correlations should be read with an effective sample size smaller than the nominal count in mind; we therefore report task-stratified correlations as the primary analysis, and the central associations remain significant at that level. Second, the RFF approximation error is measured with a finite random-feature budget (which itself approximates the RBF kernel with relative error 0.08–0.17 for the gamma values used); the simulability metrics therefore contain sampling noise that dilutes, but cannot create, the reported associations.

4 Results

4.1 Spectral criteria predict classical simulability

All five spectral criteria correlate significantly with the RFF approximation error in the predicted direction (Table 1 and Fig. 1, $n = 64$ configurations). The strongest signal comes from top-5 energy: kernels whose spectrum concentrates most of its weight in the leading eigenvalues are the easiest for RFF to reproduce ($\rho = -0.804$, $p = 1.2\times10^{-15}$; FDR $q < 10^{-14}$). Spectral entropy ($\rho = +0.674$, $p = 9.9\times10^{-10}$) and effective rank ($\rho = +0.610$, $p = 8.8 \times 10^{-8}$) show the same pattern, and all remain significant after Benjamini–Hochberg correction. Binned comparison confirms the effect: configurations below the median effective rank have mean RFF error 0.39 versus 0.82 above it, a factor of 2.1.

This result is clean in a specific sense: simulability is a property of the kernel function alone. The RFF error depends only on how well a translation-invariant classical family approximates K, and does not involve any task labels. Spectral criteria therefore answer the simulability question reliably, at negligible computational cost.

4.2 No global spectral predictor of quantum advantage

The contrast with Sec. 4.1 is sharp. In the single-task setting of exp02 ($n = 72$), spectral criteria appear to predict advantage: effective rank correlates with it at

Table 1: Spectral criteria versus RFF approximation error (exp01, $n = 64$). q: Benjamini–Hochberg FDR-adjusted.

Criterion	Spearman ρ	p	FDR q
top-5 energy	-0.804	1.2×10^{-15}	$< 10^{-13}$
spectral entropy	$+0.674$	9.9×10^{-10}	5.7×10^{-9}
effective rank	$+0.610$	8.8×10^{-8}	3.4×10^{-7}
concentration index	-0.610	8.6×10^{-8}	3.4×10^{-7}
gap ratio	-0.551	2.3×10^{-6}	7.7×10^{-6}

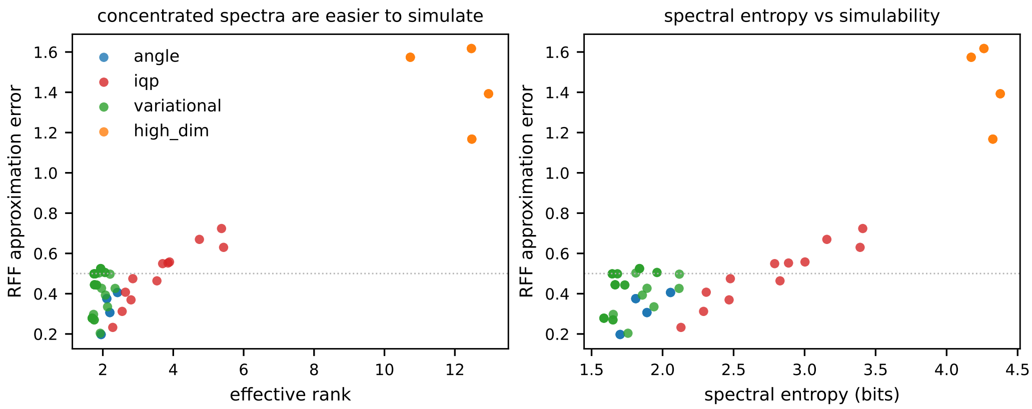

Figure 1: Spectral criteria versus classical simulability (exp01). Left: effective rank versus RFF approximation error, with the four feature-map families shown separately. Right: spectral entropy versus RFF error. Concentrated spectra are systematically easier for RFF to reproduce.

$\rho = +0.382$ ($p = 9.2 \times 10^{-4}$) and gap ratio at $\rho = -0.537$ ($p = 1.1 \times 10^{-6}$). This is exactly the kind of result that motivates spectral pre-screening of feature maps.

When we widen the setting to six tasks (exp03, $n = 216$), the global association collapses to nothing: effective rank versus advantage $\rho = -0.021$ ($p = 0.76$), spectral entropy $\rho = -0.052$ ($p = 0.45$), top-5 energy $\rho = +0.069$ ($p = 0.31$). No threshold window survives FDR correction. More strikingly, the sign is not even stable: stratified by task, effective rank correlates *negatively* with advantage on the cross_high task ($\rho = -0.628$, $p = 4.1 \times 10^{-5}$, FDR $q = 1.2 \times 10^{-4}$), *positively* on noisy_harmonic ($\rho = +0.337$, $p = 0.044$), and insignificantly on the remaining four tasks. The same criterion produces a significantly positive, a significantly negative, and several null associations within one study.

The single-task signal of exp02 was therefore a task-specific artifact, not a general law. We conclude that spectral criteria do not predict generalization advantage, and that apparent "spectral pre-screening for advantage" results from pooling within a favourable task–map combination.

4.3 Advantage window is set by classical saturation

What does predict advantage? The strongest predictor in the entire study is the *saturation of classical baselines* on the task. Across all 216 configurations, the

Table 2: Task-level summary (exp03, $n = 216$; 12 feature-map variants per task, 3 seeds). Adv: mean $\mathrm{acc}_Q - \max_B \mathrm{acc}_B$; Pos: fraction of configurations with positive advantage.

Task	mean best classical	mean Adv	Pos	task-internal $\rho_{\mathrm{Adv,diff}}$
noisy_harmonic	0.546	+0.004	38.9%	-0.507 ($q = 4.1 \times 10^{-3}$)
periodic_coupled	0.519	-0.015	27.8%	-0.431 ($q = 2.0 \times 10^{-2}$)
quantum_friendly	0.639	-0.060	13.9%	-0.367 ($q = 5.3 \times 10^{-2}$)
harmonic	0.574	-0.085	0%	-0.345 ($p = 0.040$)
cross_high	0.861	-0.225	0%	-0.380 ($q = 4.6 \times 10^{-2}$)
real_breast_cancer	0.944	-0.174	0%	-0.049 (ns)

best classical test accuracy correlates with quantum advantage at $\rho = -0.651$ ($p = 1.8 \times 10^{-27}$; FDR $q = 4.2 \times 10^{-26}$; Table 2 and Fig. 2). At the task level the pattern is monotone: the two tasks where classical baselines stay near chance (noisy_harmonic, 0.546; periodic_coupled, 0.519) are the only ones where positive-advantage configurations appear frequently (38.9% and 27.8%); the task with the highest classical baseline (real_breast_cancer, 0.944) has none.

The task-internal correlations in Table 2 matter: the association between advantage and classical difficulty holds *within* most tasks, not only across them. That is, even holding the task fixed, configurations where the classical baseline happens to be weaker (for instance due to a particular map variant) are the ones where the quantum kernel comes closer. This is consistent with a mechanism: the quantum kernel provides an alternative hypothesis space that is useful precisely when the classical hypothesis space is failing, and useless—or worse—when the classical space already solves the problem.

4.4 Design knobs and the frequency-scale law

If spectral criteria are reliable simulability indicators, which design knobs actually move the spectrum? exp04 sweeps frequency scale s, entanglement strength e, and

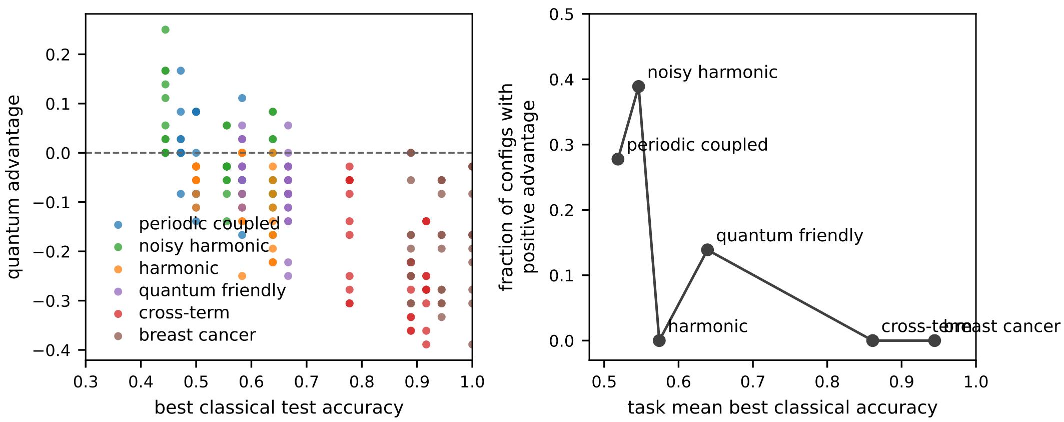

Figure 2: Quantum advantage versus the best classical test accuracy (exp03, $n = 216$). Left: every configuration, colored by task. Right: task-level means, showing the monotone decrease of positive-advantage fraction with classical difficulty.

Table 3: Design knobs versus spectral structure and simulability (exp04). Correlations are configuration-level Spearman coefficients ($n = 45$ for frequency scale and entanglement sweeps, $n = 15$ for IQP depth).

Knob	ρ vs effective rank	ρ vs RFF error	significance
frequency scale (high-dim)	+0.985	+0.884	$p < 10^{-15}$
entanglement strength (variational)	+0.093	−0.044	ns
depth (IQP)	+0.400	+0.775	rank ns; RFF $p = 7 \times 10^{-4}$

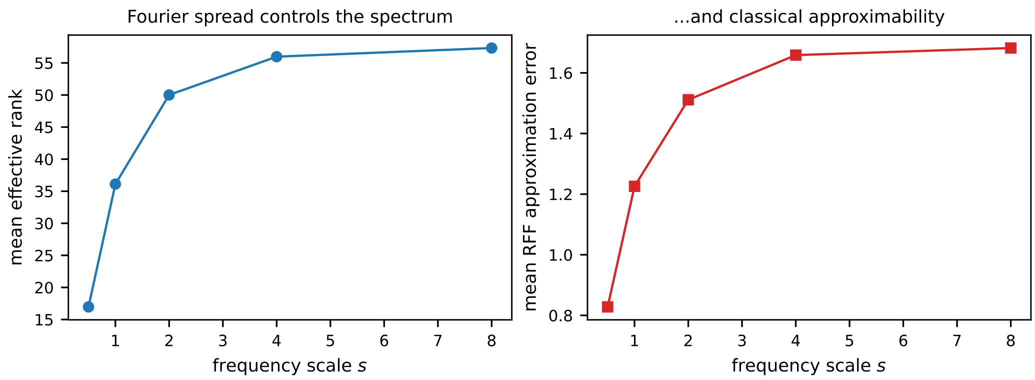

Figure 3: The frequency scale s of the high-dimensional feature map is the dominant design knob (exp04, depth 2). Left: s versus effective rank. Right: s versus RFF approximation error. Both increase monotonically with s ($\rho = +0.985$ for effective rank and $+0.884$ for RFF error, configuration level).

circuit depth d (Table 3 and Fig. 3). The frequency scale is decisive: effective rank and RFF error both increase monotonically with s (configuration-level correlations over all 45 swept settings: $\rho = +0.985$, $p = 3 \times 10^{-34}$ for effective rank; $\rho = +0.884$, $p = 9 \times 10^{-16}$ for RFF error). A larger frequency scale spreads the Fourier content of the feature map, distributing the kernel spectrum (higher entropy, higher effective rank) and making the kernel harder for RFF to approximate. Entanglement strength from 0 to 1 has essentially no effect on either quantity ($\rho = +0.09$, ns), a counterintuitive result in light of the folklore that entanglement drives concentration. For the IQP family, depth moves the RFF error ($\rho = +0.775$, $p = 7 \times 10^{-4}$) but not the effective rank ($\rho = +0.40$, ns), hinting at a mismatch between spectral concentration and classical approximability.

The second finding of this section is best stated as a counterexample to the intuitive identification of low rank with classically simulable. RFF builds an approximation in the span of random Fourier features, which by construction lives in the class of translation-invariant kernels. A quantum kernel that is simultaneously concentrated (low rank) and non-translation-invariant (IQP's quadratic phases) is therefore poorly approximated by RFF even though it looks spectrally "concentrated". Spectral criteria measure concentration, not invariance; simulability predictions require both. The same point applies to the construction of the encoding itself: data-encoding circuits built from diagonal gates only (as one of our preliminary IQP implementations was) do not leave the computational-basis state and trivially produce identity-looking kernels. We verified unitarity and encoding non-triviality with dedicated unit tests before running the study; we report this as a methodological caution for practitioners, since such errors silently invalidate spectral analyses.

4.5 Spectral criteria are data-dependent

A criterion is only useful if its value is interpretable across settings. exp05 measures how effective rank varies for a fixed feature map when the data distribution changes (six distributions: Gaussian, clustered, periodic, uniform, heavy-tailed, and a real breast-cancer slice). Coefficients of variation are large for most maps (IQP 0.38, angle 0.33, variational 0.25) and smaller for the high-frequency map (0.15 at $s = 1.5$, 0.06 at $s = 3$). Crucially, the variation is driven by the *shape* of the distribution, not by its intrinsic dimensionality: PCA intrinsic dimension correlates with effective rank only marginally ($\rho = +0.167$, $p = 0.068$). A spectral criterion computed on one dataset therefore cannot be transported to another without recalibration.

4.6 Robustness: scaling, real data, and noise

Three objections to the small-scale synthetic setting are addressed directly (Fig. 4). First, *system size*. At $n = 8$ and 12 qubits the simulability relation strengthens rather than weakens: effective rank versus RFF error reaches $\rho = +0.797$ ($p < 10^{-5}$ for both scales), compared to +0.610 at 5 qubits, and remains strong at 16 qubits ($\rho = +0.776$, $p = 8 \times 10^{-6}$). Notably, the effective rank of the angle encoding stays flat (≈ 9.5) from 8 to 16 qubits for fixed data, so in this regime we do not observe the "larger system, more concentration" trend that asymptotic arguments predict; IQP and high-dimensional maps behave similarly. The advantage– difficulty association also survives ($\rho = -0.428$, $p = 0.037$ at 8 qubits; -0.369, $p = 0.076$ at 12 qubits), and the task-internal instability of the rank–advantage sign repeats (-0.475 on cross-term, $+0.045$ on noisy-harmonic at scale).

Second, *real data*. On four standard datasets (iris, wine, breast-cancer, digits) with 8 qubits, positive quantum advantage is never observed: 0/48 configurations, with classical baselines at 0.971–1.000 and advantages in $[-0.42, -0.05]$ (Fig. 4, right). This is exactly what the classical- saturation predictor requires, and it provides external validation on data we did not construct. The simulability relation is the strongest seen so far on real data ($\rho = +0.974$, $p = 2.5 \times 10^{-31}$).

Third, *noise*. Under depolarizing noise up to $p = 0.3$ (density-matrix simulation), advantage improves slightly (-0.059 to -0.015, ns), consistent with noise acting as regularization, but never flips sign; the simulability relation remains positive ($\rho = +0.347$, $p = 0.038$). RFF errors increase with noise (0.69 to 1.81), i.e., noise makes the kernel harder to approximate classically, but the direction of all central relations is preserved. The second noise channel we test, amplitude damping, behaves differently and reveals an important boundary of spectral criteria (Table 4): damping rapidly collapses the effective rank (11.8 to 2.2, $\rho = -0.79$, $p < 10^{-5}$), and under this collapse the simulability relation *breaks down* ($\rho = +0.044$, $p = 0.80$, versus +0.347 under depolarization). Damping drives the kernel toward a constant, information-free object rather than toward a translation-invariant kernel, so spectral

Table 4: Noise-type dependence of the simulability relation (5 qubits, exp08 and exp11). ρ: Spearman correlation between effective rank and RFF error across the noise sweep; rank at max noise: mean effective rank at highest noise level.

Channel	ρ (rank vs noise)	ρ (rank vs RFF)	rank at max noise	advantage
depolarizing ($p \leq 0.3$)	-0.144 (ns)	$+0.347$ ($p = 0.04$)	9.4	$-0.059 \rightarrow -$
amplitude damping ($\gamma \leq 0.4$)	-0.789 ($p < 10^{-5}$)	$+0.044$ (ns)	2.2	$-0.059 \rightarrow -$

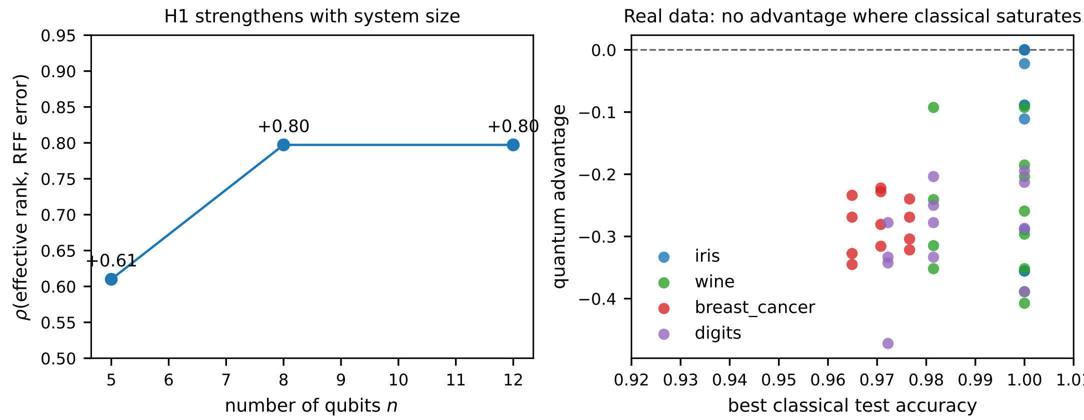

Figure 4: Scaling and real-data robustness. Left: the simulability relation (ρ of effective rank vs. RFF error) strengthens with system size. Right: on four real datasets the classical baseline is near-saturated and quantum advantage is never positive.

concentration no longer indicates classical approximability. The two-channel comparison shows that whether a spectral criterion remains a valid simulability proxy depends on the noise type, not merely on the presence of noise.

4.7 Two matching criteria fail to explain residual advantage

Is there any task–map matching quantity that predicts advantage beyond classical difficulty? We test the two most natural candidates on all 216 exp03 configurations (Fig. 5). The spectral spread ratio $M_1 = \mathrm{rank}_Q/\mathrm{rank}_{\mathrm{RBF}}$ captures whether the map broadens the spectrum relative to the classical benchmark kernel; it correlates with advantage only marginally ($\rho = +0.125$, $p = 0.067$) and not at all after controlling for classical difficulty (residual $\rho = -0.104$, $p = 0.127$). Task-level spread ratios do not separate tasks with positive from negative rank–advantage correlations ($\rho = -0.03$ across six tasks, $p = 0.96$).

The second candidate, centered kernel-target alignment $\mathrm{TA}(K, yy^\top)$ [7], is the standard quantity used to argue that a kernel "fits" a task. At face value TA predicts advantage strongly ($\rho = -0.417$, $p = 1.7 \times 10^{-10}$), but this association is entirely explained by task difficulty: after rank-space residualization on the best classical accuracy, the remaining correlation is $\rho = +0.001$ ($p = 0.99$). Alignment correlates with easy tasks; easy tasks correlate with saturated classical baselines; saturated classical baselines correlate with absent quantum advantage. The apparent predictive power of TA is thus a chain of difficulty correlations rather than an

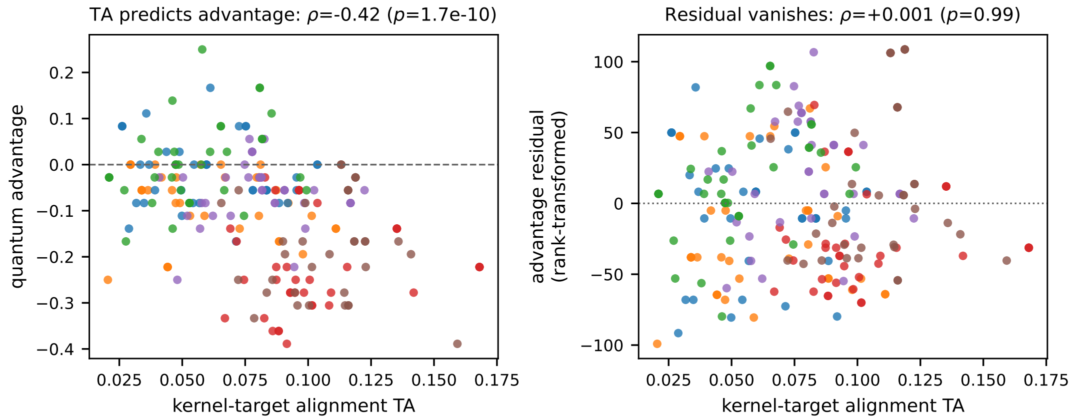

Figure 5: Kernel-target alignment and quantum advantage (exp03, $n = 216$). Left: TA correlates with advantage globally ($\rho = -0.42$). Right: after rank-space residualization on the best classical baseline, the association vanishes ($\rho = +0.001$). The full predictive power of TA is absorbed by task difficulty.

independent signal. This is a clean empirical boundary for the alignment-based intuition: in our setting, no simple spectral or alignment criterion carries information about advantage that the classical baseline does not already carry.

5 Discussion

Two questions are routinely conflated in the quantum-kernel literature: whether a kernel is classically simulable, and whether it confers an advantage. Our results draw a sharp line between them. Spectral criteria are excellent instruments for the first question alone. They are cheap, task-independent, and their prediction of RFF/Nyström approximability is statistically strong and directionally consistent. They are not instruments for the second. Across tasks and within tasks, the same spectral measures produce positive, negative, and null associations with advantage, and the global correlation is indistinguishable from zero. Any protocol that prescreens feature maps by spectral criteria with the goal of predicting generalization gain lacks empirical basis.

The dominant observable predictor of advantage is instead the classical baseline's saturation. This has an immediate practical reading. The quantum kernel competes with classical alternatives; when the classical hypothesis space already contains an adequate solution, an exotic alternative has no room to win, however expressive it is. Conversely, tasks that classical kernel methods find hard—here, periodic coupling and noisy harmonic structure near chance-level classical accuracy—are exactly where quantum kernels can match or exceed them. This is consistent with the spirit of dequantization results: large claims about quantum advantage are most fragile precisely when classical baselines are strong [4, 5].

Our findings yield three concrete recommendations for the evaluation of quantum kernels. First, classical baselines must be tuned (validation-based kernel width at least) and must include a spectrum of families (RBF, RFF, linear, polynomial); untuned baselines inflate apparent advantage. Second, task difficulty must be reported alongside advantage; an advantage of +0.03 on a task where classical baselines reach 0.54 is qualitatively different from +0.03 on a task where they reach 0.94, and only the former is informative about the quantum hypothesis space. Third, spectral pre-screening should be used for simulability diagnostics only; statements about advantage require the task-resolved protocol used here.

The failure of both matching criteria deserves emphasis, because it constrains future work. Our result on kernel-target alignment is, in effect, a null result about a popular intuition: when a quantum kernel appears well aligned with the labels, that appearance is confounded with an easy task and a strong classical baseline. This does not contradict recent findings that dataset choice dominates variance and that typical quantum feature-map spectra are poorly shaped compared with RBF [10], or that data-adaptive qubit budgets can predict collapse [11]; it shows that these observations, and any alignment-based pre-screening, should be interpreted in a framework where classical baselines are the reference point. It also echoes the early argument of Kübler et al. [7] that the inductive-bias question must be asked at the level of the target function, not the kernel alone.

5.1 Limitations

Several limitations bound the scope of these conclusions. Although we extend the main analysis to 8–16 qubits, real datasets, and depolarizing noise, the accessible regime remains small relative to the asymptotics that classical-simulability theorems concern; in particular, entanglement-scaling effects at tens of qubits are untested. The noise model is limited to global depolarizing and single-qubit amplitude-damping channels; correlated noise and measurement noise are not included. Real-data experiments use PCA preprocessing to eight features and a single classifier (kernel SVM) with fixed shrinkage. The task family, while deliberately diverse, still includes only one quantum-agnostic real dataset family, and the matching-criterion analysis is confined to the six tasks of exp03. The RFF family approximates translation-invariant kernels; simulability statements should be read relative to that reference class. Finally, all simulations are noiseless except exp08, and classical surrogates beyond RFF and Nyström (e.g., tensor-network contractions) are not considered.

6 Conclusion

We asked what spectral criteria of quantum feature-map kernels actually predict, and answered it with a task-resolved numerical protocol built on a self-contained simulator. The answer is asymmetric. Spectral criteria reliably predict classical simulability: all five measures correlate significantly with RFF approximation error, with top-5 energy reaching $\rho = -0.80$. They do not predict quantum advantage: the global association vanishes, and even the sign is unstable across tasks. The single robust predictor of advantage in our study is the saturation of the classical baseline ($\rho = -0.65$, FDR $q = 4 \times 10^{-26}$), with positive-advantage configurations appearing almost exclusively where classical kernels remain near chance. We also identified the frequency scale of the feature map as the dominant design knob controlling both spectral spread and classical approximability, and established that spectral criteria are strongly data-dependent.

For practitioners, the operational takeaway is short: use spectral criteria to estimate how hard a quantum kernel is to replace classically, and use strong, tuned classical baselines on tasks above classical saturation to measure whether it is worth replacing the classical model at all.

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Data and code availability

All simulation code, analysis scripts, and raw result data (CSV and figure generation scripts) are publicly available at github.com/python123flask/quantum-kernel-spectral. All experiments are reproducible with NumPy, SciPy, scikit-learn, and matplotlib only.

References

[1] Maria Schuld and Nathan Killoran. "Quantum machine learning in feature Hilbert spaces". Physical Review Letters **122**, 040504 (2019).

[2] Vojtěch Havlíček, Antonio D. Córcoles, Kristan Temme, Aram W. Harrow, Abhinav Kandala, Jerry M. Chow, and Jay M. Gambetta. "Supervised learning with quantum-enhanced feature spaces". Nature **567**, 209–212 (2019). url: `https://arxiv.org/abs/1804.11326`.

[3] Supanut Thanasilp, Samson Wang, Marco Cerezo, and Zoë Holmes. "Exponential concentration in quantum kernel methods". Nature Communications **15**, 5200 (2024).

[4] Sunghun Shin et al. "Dequantizing quantum machine learning models using tensor networks". Physical Review Research **6**, 023218 (2024).

[5] Ryan Sweke et al. "Potential and limitations of random fourier features for dequantizing variational quantum machine learning". Quantum **9**, 1640 (2025).

[6] Salvatore Thabet et al. "When quantum and classical models disagree: learning with random fourier features". npj Quantum Information (2026).

[7] Jonas M. Kübler, Simon Buchholz, and Bernhard Schölkopf. "The inductive bias of quantum kernels". In Advances in Neural Information Processing Systems 34. (2021). url: `https://arxiv.org/abs/2106.03747`.

[8] Valentin Heyraud, Zejian Li, Zacharie Denis, Alexandre Le Boité, and Cristiano Ciuti. "Noisy quantum kernel machines". Physical Review A **106**, 052421 (2022).
[9] Claudia Zendejas-Morales, Debashis Saikia, and Utkarsh Singh. "Local and multi-scale strategies to mitigate exponential concentration in quantum kernels" (2026). url: `https://arxiv.org/abs/2602.16097`.
[10] Siavash Kakavand, Christoph Strohmeyer, and Michael Schlotter. "Benchmarking quantum kernel support vector machines against classical baselines on tabular data: A rigorous empirical study with hardware validation" (2026). url: `https://arxiv.org/abs/2604.18837`.
[11] Ana Paula Appel. "Fractal dimension predicts quantum kernel collapse in angle-encoded data" (2026). url: `https://arxiv.org/abs/2609.00475`.
[12] Erik M. Åsgrim and Stefano Markidis. "Quantum kernels are spectral tensor networks" (2026). url: `https://arxiv.org/abs/2606.20402`.
[13] Ali Rahimi and Benjamin Recht. "Random features for large-scale kernel machines". In Advances in Neural Information Processing Systems 20. Pages 1177–1184. (2007). url: `https://proceedings.neurips.cc/paper/2007/hash/013a006f03dbc5392effeb8f18fda755-Abstract.html`.